

OFFLINE *learn once, then audit-calibrate*

$\mathcal{D}_{\text{tr}}, \mathcal{D}_{\text{val}}$

fit $\theta, \tilde{\theta}$

FNO dynamics model

$$\hat{u}_{t+1} = \hat{G}_{\theta}(u_t, c_t, f_t; \xi_{\text{obs}}) \quad (8)$$

$$c_t = \sum_{k=1}^m a_{t,k} \psi_k, \quad \xi_{\text{obs}} = (\nu, u_L, u_R)$$

Plug-in uncertainty scale

$$\sigma_{\theta}(u, a) = \text{Smooth}(|\hat{G}_{\theta} - \tilde{G}_{\tilde{\theta}}|(u, a)) + \tau_0 \quad (10)$$

residual spread of the twin models

Split-conformal calibration

$$z_i = (u_i, c_i, f_i; \xi_i^{\text{obs}}), \quad r_i = y_i - \hat{G}_{\theta}(z_i)$$

$$S_i^{(g)} = \|r_i \odot \sigma_i^{-1}\|_g, \quad q_g = S_{(\lceil (n_{\text{cal}}+1)(1-\alpha) \rceil)}^{(g)} \quad (12)$$

z_i is one complete operator input

calibration only, no FNO retraining

frozen $\hat{G}_{\theta}, \sigma_{\theta}, q_g$

ONLINE *robust receding-horizon control*

u_t
 $f_{t:t+H-1}^{\text{ext}}$
 ξ_{obs}

Calibrated ambiguity set

$$\mathcal{U}_g(u, a) = \{\hat{G}_{\theta}(u, a) + \Delta : \|\Delta \odot \sigma_{\theta}^{-1}\|_g \leq q_g\} \quad (14)$$

$g = 2$: ellipsoid | $g = \infty$: simultaneous box

Support function

$$h_{\mathcal{U}_g}(\lambda) = \sup_{\Delta \in \mathcal{U}_g} \langle \lambda, \Delta \rangle_n$$

$$h_{\mathcal{U}_2}(\lambda) = q_2 \|\lambda \odot \sigma\|_{2,n}$$

rollout queries (u_h, a_h) , adjoint λ

Robust receding-horizon control

$$a_{t+H-1}^* = \arg \min_a \max_{\Delta_h \in \mathcal{U}_g(u_h, a_h)}$$

$$\left[\sum_{h=0}^{H-1} \ell(u_h, a_h) + \ell_T(u_H) \right] \quad (17)$$

apply the first action a_t^*

Forced PDE plant

$$\partial_t u + u \partial_x u = \nu \partial_{xx} u + f_{\text{ext}}(x, t) + g_{\text{act}} c_t(x)$$

g_{act} latent: never an input to $\hat{G}_{\theta}$

u_{t+1} : observe, update horizon, and replan